EE421 - Geometrical Optics Project #2 Spectrometer Design in Zemax OpticStudio

Selin Nisa Akgöl
Electrical-Electronics Engineering
Abdullah Gül University
Kayseri, Türkiye
selinnisa.akgol@agu.edu.tr

I. Introduction

Lurie-Houghton or Houghton Telescope is a **catadioptric type** telescope that utilizes a two-lens corrector, also referred to as the Houghton Corrector, to correct the image and a spherical mirror at the back to magnify and focus [1] as well as a focusing mirror, and a redirecting mirror.

II. Telescope Summary

The telescope is built according to Rick Scott's design parameters given in Table I.

Description	Parameter	Original (cm)
First corrector lens (positive, BK-7)	R1	184.4
	T1	1.54
	R2	-528.3
Air space between corrector lenses	T2	0.28
Second corrector lens (negative, BK-7)	R3	-184.4
	T3	0.95
	R4	528.3
Air space: corrector to primary	T4	95.3
Spherical primary mirror	R5	-228.0
Air Space: primary to secondary	T5	87.9
Flat Diagonal Mirror	R6	0

TABLE I
Lurie-Houghton Design Parameters [2]

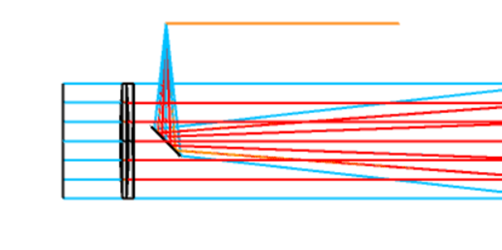


Fig. 1. Complete Lurie-Houghton Telescope

Optical Parameters:

- **Effective focal length (EFL)** = 114.4978cm
- **f-number (F#)** = 4,507508
- **Location of the image plane** is where the telescope focuses the image from infinity. Refer to Figure 1.

III. Spectrometer Design

A proper spectrometer must have an entrance slit to narrow down the number of entering rays, a collimating lens if the rays are not parallel to each other, a dispersive element to separate the light into different wavelengths, an imaging lens to refocus the now separated and colored rays, and finally a detector plane to detect the various wavelengths.

A. Overall Layout

- **Entrance slit:** Rectangular aperture with X=0.002cm Y=0.3cm.

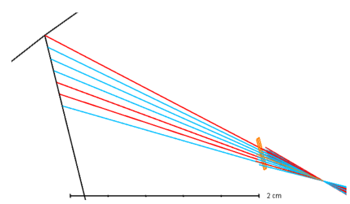


Fig. 2. Entrance Slit

The position of the slit also creates a difference when it comes to how many of the rays pass through to the spectrometer system, as well as the width of the slit. If the slit is closer to where the telescope focuses the light, aka the start of the spectrometer, more rays pass through. The point where rays are separated enough to let fewer of them pass improves the resolution of the image on the detector plane.

- **Collimating lens:** A plano-convex focusing lens was implemented with the use of its focal length formula to decide on the radius of this element.

$$R = (n_{BK7} - 1)f = 0.5168f \tag{1}$$

Where f is chosen to be 5cm, as it does not depend on the system before. Therefore, the radius R is 2.584cm. The polarity is changed according to the system's requirements.

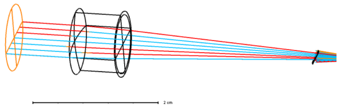


Fig. 3. Collimator Lens

- **Dispersive element:** This step is achieved by inserting a surface after the focusing lens and changing the surface type to 'Diffraction Grating'. In Zemax, the diffraction grating is simulated with:

$$n_2 sin\theta_2 - n_1 sin\theta_1 = M\lambda T \quad (2)$$

Where M is the diffraction order, λ is the wavelength, T (line/μm) is the grating period [3].

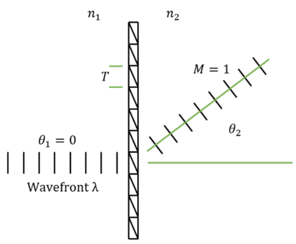


Fig. 4. Diffraction grating sketch

Incident rays are entering the diffraction grating with an angle of 0 or very close to that; therefore, $n_1 sin\theta_1$ term becomes zero. To make the understanding simpler, this study kept the initial setting in Zemax for diffraction order and diffraction period 1 so that dispersing only depends on wavelength.

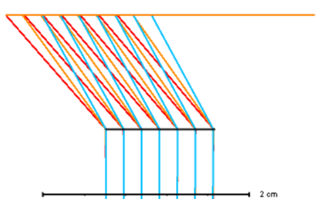


Fig. 5. Diffraction grating

- **Imaging lens:** Another focusing lens is used but this time with changing the surface type to 'Paraxial' and setting the focal length 1cm.

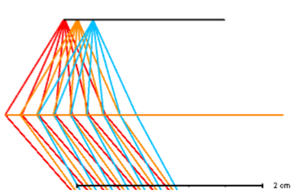


Fig. 6. Focusing Lens

- **Detector plane:** The detector plane is the image plane, and it is set exactly 1cm away from the focusing lens surface.

The spectrometer in full is:

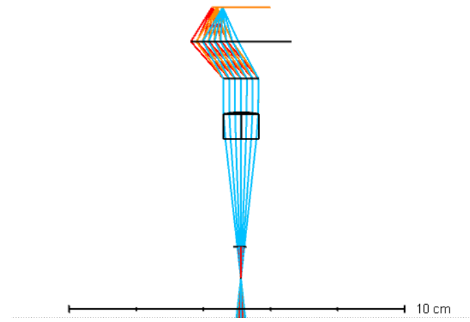


Fig. 7. Spectrometer in Zemax

Complete system is:

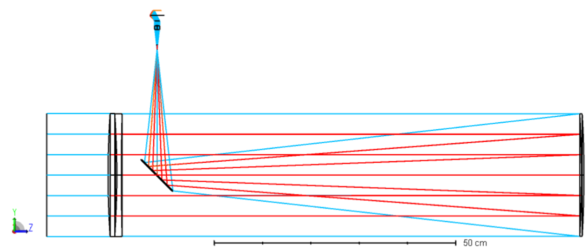


Fig. 8. The complete system, Lurie-Houghton telescope & spectrometer

IV. Optical Analysis

A. Wavelength Definition

The system is analyzed at seven visible wavelengths:

- 400 nm
- 425 nm
- 470 nm
- 550 nm
- 600 nm
- 630 nm
- 675 nm

B. Ray Diagram

The ray diagram of the spectrometer is shown in Figure 9. Different colors of the light are clearly separated and focused on the detector plane.

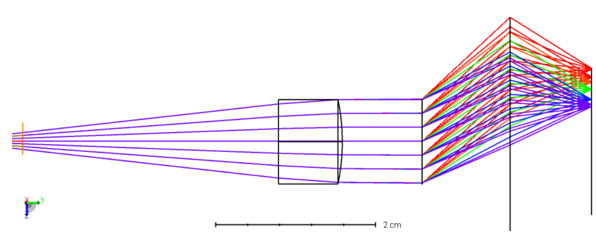


Fig. 9. Ray diagram of the spectrometer

C. Spot Diagram

The spots that appear on the detector plane can be seen in Figure 10. A clear wavelength separation is visible for the wavelengths with which the simulation is done.

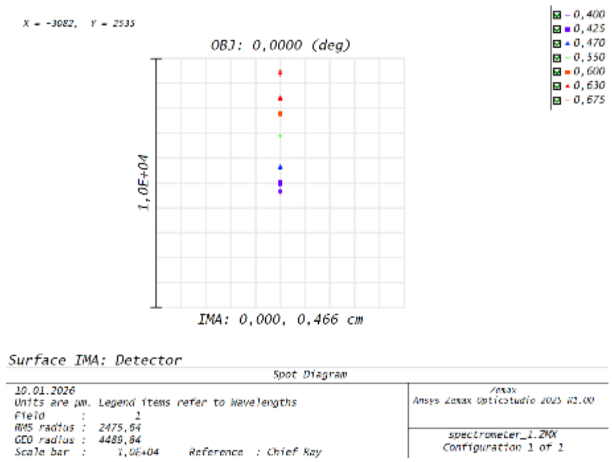
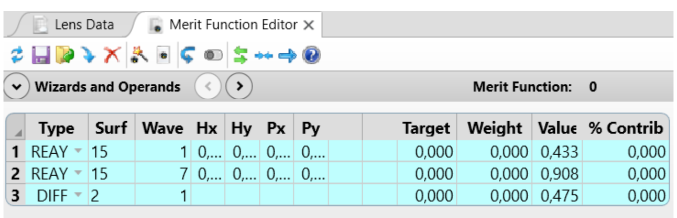


Fig. 10. Spot diagram at the detector (image) plane

D. Spectral Image

The 'Merit Function Editor' window that is accessed from the 'Automatic Optimization' in 'Optimize' ribbon has the function to tell the distance between two different wavelengths' chief ray's position on the image plane, or on any surface.



	Type	Surf	Wave	Hx	Hy	Px	Py	Target	Weight	Value	% Contrib
1	REAY	15	1	0,...	0,...	0,...	0,...	0,000	0,000	0,433	0,000
2	REAY	15	7	0,...	0,...	0,...	0,...	0,000	0,000	0,908	0,000
3	DIFF	2	1					0,000	0,000	0,475	0,000

Fig. 11. Merit Function Editor Window

From Figure 11, it can be seen that the difference between the 400nm and 675nm waves is 0.475cm. Assuming the detector has $10\mu m$ pixels lined up as the industrial standard for CCDs, the spatial resolution of this system can be calculated with:

$$\frac{\lambda_{max} - \lambda_{min}}{x} p = \frac{675nm - 400nm}{0.475cm} 10\mu m = 0.58nm/pixel \tag{3}$$

where p is the pixel size and x is the spatial difference between the maximum and minimum wavelengths.

V. Performance Discussion

The system overall has poor resolution as the close wavelengths overlap and make distinction harder on the detector plane, but regardless, the wavelengths are clearly separated as seen in the Figure 9.

Slit width decides how many of the rays pass from the entrance. In this study, the used ray settings were 'number of rays:7', 'Grid' in the 3D layout settings. Not many rays are generated in this format, so only a single line of grip passes through, and the width of the slit does not change anything within a range. Of course, in real life smaller the width, the better the resolution. The height (the longer side) of the slit also affects the resolution greatly. It can be set to reduce the amount of rays that pass through so that the spots are more focused on the detector plane.

A. Design Limitations

The design requirements were that no complex elements were necessary. The simplicity of the lenses and other optical elements caused optical aberrations, which in turn messed with the resolution. Another aspect that affects the resolution is the height of the slit, with the reasons mentioned above. Reducing the height lets fewer rays be inflicted on a single pixel, so that there is less 'spillage' from the adjacent wavelength.

VI. Conclusion

References

[1] W. Contributors, "Ray transfer matrix analysis." *Wikipedia.*, (2023, March 28). [Online]. Available: https://en.wikipedia.org/wiki/Ray_transfer_matrix_analysis

[2] R. Scott, "Lurie-houghton telescope design with a comparison to the newtonian telescope," *Naturalimagesgallery.com*, (2002, December 23). [Online]. Available: https://naturalimagesgallery.com/atm/lh_scope/lh_design_article/lh_design.html

[3] N.-H. Kim, "How diffractive surfaces are modeled in opticstudio." [Online]. Available: https://optics.ansys.com/hc/en-us/articles/42661739263251-How-diffractive-surfaces-are-modeled-in-OpticStudio

Zemax Archive Files